UNIFY LEARN

Product Documentation

Guided 12-week paths, XP, streaks and notes that fit your courses.

Version 1.26.7 · 21 October 2026

Mobile-first learning platform for the Faculty of Engineering, Lagos State University.
Covers: current features · roles & journeys · notifications · sessions & security ·
design system · architecture · pilot readiness · multi-tenant roadmap · change log v1.12–v1.26.7.

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1 · Product overview

Unify Learn is a mobile-first web application that turns lecture material into guided, interactive weekly notes for engineering students. Authors publish versioned topics; students read, attempt quizzes, earn XP and keep streaks; admins run the whole platform from an oversight dashboard.

- **Frontend:** Vite 5 + React 18 + TypeScript 5, served on Vercel (`unify-virid.vercel.app`), route-level code-split (main chunk 190 KB raw / 52 KB gzip).
- **Backend:** Express + TypeScript + Prisma on Render (`unify-api-z4zm`).
- **Data & auth:** Supabase Postgres + Supabase Auth (email + Google OAuth).
- **Email:** Brevo (SMTP for auth mail, API for note/announcement mail from `EMAIL_FROM` — must be a verified Brevo sender).
- **Push:** Web Push (VAPID) to browsers, third notify leg.
- **Content seed:** 242 mined courses (full catalog verified), 90 legacy weeks converted to topic-level notes.

- **Offline:** installable PWA (InstallBar + update-ready dot), write queue, per-week Save offline (SW v8).

2 · Roles & access matrix

Capability	Student	Lecturer	Collaborator	Admin (role)
Learn paths (My Courses, Explore, reader)	Yes (enrolled only)	No	No	No (reads via All content)
XP, streaks, quizzes recorded	Yes	No	No	No
Studio authoring + preview	No	Yes	Yes (assigned courses only, server-enforced)	Yes
Level-scoped note browser	—	Teaching level	Contributing level	All content
Admin panel (users, courses, analytics)	No	No	No	Yes (role or legacy flag)
Role assignment	Immutable after onboarding; changed only by an admin (recipient is notified).			

The bootstrap `unify.admin` account is flag-based (`collaborator` + `is_admin`), so the flag permanently counts as admin — the main account can never be locked out. Other admins inherit full rights via the `admin` role.

3 · Student journey

1. **Auth:** email sign-up (verification mail), sign-in, **Continue with Google** (live, PKCE), forgot-password reset, resend verification. Remember-me is opt-in.
2. **Onboarding (8 steps):** role → name → university → faculty → department → level → courses (own level + active semester) → graduation target.
3. **Dashboard:** greeting, daily line, XP / streak / target / courses, Continue-Learning resume card (exact course·week·topic·lecture), quiz stats, enrolled courses as a 2-column card grid (lecturer names, one-tap elective removal).
4. **My Courses:** enrolled courses with week counts. **Explore:** enroll-only catalog for own level + active semester (single bulk call), debounced search across code/title/alias and both semesters.
5. **Reader:** story hero, segmented chapter bar, topics, recalls, quizzes, read-aloud (device voice with speed control + Nigerian `en-NG` preference), Focus + zoom controls, exact resume tracking.
6. **Notifications:** push-driven bell badge + list page; unread auto-opens on fresh entry. Email + push per profile prefs.

7. **Profile:** inline edit (any-size avatar upload with server-side magic-byte validation), email change + **verify chip with Resend**, notify prefs, appearance (Classic/Story, light/dark, push toggle), version footer, guarded logout.
8. **XP perks:** Save PDF unlocks at 300 XP; share links (8/16/24 h) for authors/admins; streaks count consecutive active days.

4 · Author journey

1. **Onboarding (lecturer 6 steps / collaborator 3):** both pick the level they teach / contribute to; courses selected enroll as *teaching*.
2. **Studio:** three methods — AI generate (Gemini, rate-limited, async chunked jobs), manual visual builder (no raw JSON; Word-like editing, tap-to-insert templates, full visual equation builder with zero LaTeX typing), external-AI format pack. Review summaries, limits and a score overlay guide publication.
3. **Versioning:** every publish stacks v1, v2, v3… per topic; occupied weeks/topics return 409 with Replace / Add-as-new / Cancel (nothing overwrites silently); authors delete own versions, admins any. Reader shows version badges + picker (latest version only in lists).
4. **Lecture structure:** weeks hold Lecture 1/2/3; Studio/NoteBuilder carry lecture pickers; AI prompt emits lecture groups.
5. **Dashboard:** Notes / Courses / Students / Done stats, own published list with preview links, level-scoped Browse entry, My contributions view.
6. **Sharing:** expiring `/s/:token` links with WhatsApp/Telegram unfurl cards, per-link OG authors, week author badges; live-links index (create/revoke) fixed in v1.11.1.
7. **Narrator audio:** optional narrator audio track per topic (v1.21).

5 · Admin journey

- **Oversight dashboard:** platform totals (users, courses, weeks, topics), latest signups, latest published notes. Main admin (role) gets the panel link; flag admins get view-only notes.
- **Admin panel (sidebar modules):** Analytics (roles donut, 14-day signups/notes/XP charts), AI models, Universities, Courses per level (search), Academic session + promote-all, Announce (in-app + email + push), Users & roles (searchable assignment, invite, role/level edit, self-lock), approval flow for contributors, avatars/tables, error log + support config.
- **All content browser:** every course → weeks → topic/version chips with read-only preview links.
- **Academic control:** active-semester toggle (students only ever see that semester), one-click promotion (500 Level graduates, enrollments reset), waiting room (v1.16).
- **Capacity governance:** server-side cap gate + audit UI (v1.18/v1.19) — quotas are enforced with a visible audit trail, not silently.

- **Safety:** own role locked (UI + 403 server-side), self-demotion blocked, every destructive/broadcast action uses the designed confirm sheet.

6 · Notifications (in-app · email · push)

Event	In-app bell	Email (Brevo)	Push
New note published	Enrolled students	Confirmed + opted-in takers (API, fan-out paginated past 200/1000 caps)	Subscribed takers
Admin announcement	All users	All email-confirmed users	All subscribed
Welcome (first onboarding)	New user	— (auth mail covers it)	—
Role promoted / demoted	Recipient	—	Recipient

Auth mail (signup verify, invite, password reset) uses three branded Supabase templates; everything else stays on Supabase defaults. Note/announcement mail goes through the Brevo API from Render's `EMAIL_FROM` — **the from-address must be a verified Brevo sender or every send fails** `brevo 400` (see SETUP.md). Profile shows an email-verify chip with a Resend action. Push needs VAPID keys on Render and per-device opt-in from Profile → Appearance.

7 · Sessions, security & XP economy

- **Independent per-tab sessions** (sessionStorage); Remember-me restores tokens across restarts (localStorage, opt-in).
- **30-minute sliding inactivity TTL** persisted across app close; expiry signs out with an explanatory message. Applies to remembered sessions too.
- **XP:** +10 per completed topic, students only (server-enforced; staff reads record nothing). Quizzes record stats, award no XP. Streaks count consecutive active days.
- **Guards:** enrolled-only learning (fail-open on flakes), role gates on every route, admin-only APIs via `requireAdminUser`, collaborator course scoping server-side (403 outside assigned courses), no 4xx retries, rate limits, server cap gate + audit, error visibility (global toasts), log-only mode without keys.

8 · Reader: story notes, math & quizzes

- **Chrome:** dark story hero (eyebrow, serif title, outcome box, chips incl. XP earned), segmented chapter bar (done/current, tap-to-jump), scroll-reveal story beats, Back/Next auto-scroll honoring the return trail, back-to-top landing on the topic head, print stylesheet.

- **Focus + zoom (v1.25–v1.26.3):** one floating bottom-right pill, rendered only on note readers. Focus hides the top bar + bottom tabs (chrome hides only on reader routes — never leaks elsewhere); desktop zoom-in only, 100–150% in 10% steps, scales *only* the content column; first zoom-in enables Focus, the % button resets both. Focus and zoom persist per device and are retained when switching notes.
- **Math (never-raw, v1.26.4):** one MathJax pipeline (`Formula` / `MathText`). Formulas show readable glyph math from first paint and only reveal real typeset output after MathJax succeeds — typeset failures (blocked CDN, malformed TeX, hung font fetch) fall back to glyphs with a hard 6-second deadline, and every successful typeset is **saved per-equation** (localStorage) so previews and notes render instantly offline. Raw `\frac…` source can never appear on screen or in a PDF. The Studio equation field keeps a live RENDERED preview box.
- **Interaction:** tap-to-reveal recall deck with Got-it / Review-again self-grading and progress; key phrases highlighted via `<mark>`; designed fill-in-the-gap fields; read-aloud mode (device voice, speed control, Nigerian voice preference, per-device memory, Drop voice picker).
- **End-of-week exam card:** progress dots, submit-at-end scoring (no answer leakage), instant verdicts with author feedback, topic-ref review chips, mascot pass/fail score screen with exact retry list; short banks auto-repair to 10 questions.
- **Save PDF (300 XP):** prints the rendered reader — controls/nav/install bar hidden, reader zoom reset to 100%, cards unbroken; math prints typeset or as glyphs.
- **Offline:** per-week Save offline into a version-proof content cache; progress/quiz/resume writes bank locally and flush on reconnect.
- **Delight:** XP bursts on completion (+10 XP) and week-complete celebration; reduced-motion respected throughout.

9 · Design system

- **Classic (default):** the previous look — white surfaces, Nunito, emerald, Duolingo-style retention mechanics (chunky targets, XP bursts, celebration moments).
- **Story (opt-in):** the MEE352 foundation — warm paper, Playfair Display + DM Sans, editorial hero. Adopted from the original `mee352-week2` source.
- **Dark mode** re-skins either design; persisted, no flash on return. Toggle lives in Profile → Appearance with the design switch and push toggle.
- **Overlay dropdowns:** every selector uses the shared `Drop` component — never a native `<select>`.
- **Brand:** Unify wordmark lockup app-wide (v1.13); designed confirmations everywhere (zero browser popups); role-aware skeletons that match what loads; global error toasts mounted at app root.
- Both themes are co-equal official themes (“Classic / Play” and “Story / Editorial”), persisted per device — see DEVELOPER-REPORT § brand direction for the ruling on the generated brand kit.

10 · Versioning scheme

Semantic versioning `MAJOR.MINOR.PATCH`, shown in Profile (currently **v1.26.7**):

- **MAJOR** — platform generation (1 = current app).
- **MINOR** — feature batch (new screens, flows, capabilities).
- **PATCH** — bugfix/hotfix push with no new surface.

> **Rule:** every push bumps `APP_VERSION` and carries the version in the commit message; the frontend mirror is re-synced to `DIBBLS/Unify` after every push (Vercel auto-builds).

11 · Architecture & data model

Layer	Tech	Notes
Web	Vite 5 · React 18 · TS 5 · Router 6 · MathJax	Static on Vercel; route-level code-splitting; offline worker (SW v8, push-enabled, content cache, narrator audio + media cache-first)
API	Express 4 · TS · Prisma (DDL only)	Render; `prisma migrate deploy` on every build (migrations 0016–0023 applied)
Data/Auth	Supabase Postgres + Auth	Service-role reads via API; pooler :6543, direct :5432 for migrate
Mail/Push/AI	Brevo · Web Push (VAPID) · Gemini	All log-and-skip without keys

Core tables

`profiles` (role, level, is_admin, prefs) · `enrollments` (taking/teaching) · `courses` + `course_levels` + `course_aliases` · `weeks` (shells, lecture-keyed) · `topic_notes` (versioned content) · `topic_progress` · `xp_events` · `quiz_attempts` · `resume_state` · `notifications` · `push_subscriptions` · `app_settings` (semester) · `ai_models` · `universities` · `class_slots` (timetable + roster)

API map (v1)

Auth/profile: `/me` · `PUT /me` · `/avatar` · `/onboarding` · `/settings` · `/stats` · `/resume` · enrollments (+repair) · universities · courses (+weeks counts, aliases search) · course weeks/week detail (+lecture paths) · progress · quiz/attempt · authored · author/stats · notes get/delete · notifications (+read) · push subscribe/unsubscribe · publish ×2 (paginated fan-out) · share create/resolve/revoke/mine · classes timetable + roster · error log + support config · admin: stats, activity, trends, content, announce (in-app + email + push), users, promote, courses, unis, semester, models, caps audit.

12 · Pilot readiness: the 500-user verdict

Verdict: YES for a 500-tester pilot — with two paid switches and one code fix (done). 500 registered testers using the app normally will fit; 500 *simultaneous* users on the free tier will not.

Layer	Free-tier limit	500-tester pilot
Vercel (static frontend + CDN)	Generous bandwidth	Fine as-is.
Render free API (512 MB, shared CPU, sleeps)	~20–50 concurrent, 30–60 s cold starts	**Upgrade to Starter ($7/mo)**: no sleep, 1 GB RAM.
Supabase free (500 MB DB, 5 GB bandwidth, pooled conns)	Fine for hundreds of MAU	Fine as-is; keep the :6543 pooler. Pro ($25) only if you outgrow it.
App N+1 (Explore fired ~160 requests/visit)	Would melt Render in a pilot	**Fixed (v1.6.0):** bulk week counts, one request per catalog.

Pilot runbook

1. Render Starter; confirm `/healthz` warm; set VAPID + Brevo + Gemini keys (and verify the Brevo sender).
2. Seed + promote: run `seed.sql` (242-course catalog), verify catalog, set active semester, create the admin.
3. Invite in waves (e.g. 5 × 100) with Supabase Auth logs + Render metrics open.
4. Watch: p95 latency, 5xx rate, Supabase connection count, Brevo daily quota (300/day free).
5. Success bar: signup → onboarding → first week → first quiz → Day-2 return for the cohort.

Do not pilot on Render free: cold starts alone will read as "the app is broken" to testers, and a class-wide signup surge will cascade into timeouts.

13 · Multi-tenant production plan (many universities)

Target: one deployment serving LASU + other universities, each isolated, each brandable, centrally billed. Recommended path — **single database, tenant-scoped rows** (not a DB per school):

1. **Tenant key:** add `tenants` table (subdomain, name, logo, colors, sender email); add `tenant_id` to profiles, courses, weeks, notes, enrollments, settings. Every query scopes by tenant (one middleware + RLS policies).
2. **Routing:** `lasu.unify.learn`, `unilag.unify.learn` → tenant resolver; keep path fallback (`?t=`) for testing.
3. **Auth:** email-domain rules per tenant (e.g. `@lasu.edu.ng` auto-joins LASU); invites carry tenant; one user can belong to two tenants (student here, lecturer there). *Proposed, not built: per-university*

allowed-email-domains policy (works with both email and Google paths).

4. **Content:** shared starter catalog cloned per tenant on provisioning; each school curates its own weeks/notes/semesters.
5. **Admin:** tenant admins (current panel, scoped) + super-admins (cross-tenant analytics, provisioning, billing).
6. **Mail/push:** per-tenant sender identities (Brevo subaccounts/domains), per-tenant VAPID subjects.
7. **Infra:** Render Standard+ (or autoscaling host), Supabase Pro (backups, connection headroom, PITR), read-replica if analytics bite; per-tenant rate limits + quotas.
8. **Migration:** backfill all existing rows to the LASU tenant; dual-run; cut over per school. Estimated: 3–5 focused batches (schema → scoping → routing → admin → infra).

14 · Roadmap ("later")

- Multi-tenant universities (plan above; allowed-email-domains policy first).
- Full offline packs (download a course, learn on zero data) — per-week Save offline ships now; course-level packs are the next step.
- Native-feel install prompts + iOS push parity notes (InstallBar ships; iOS specifics pending).
- AI tutor over mastered topics; spaced-repetition scheduling from recall grades.
- Leaderboards, study groups, lecturer analytics per topic.
- Certificates + transcript export for departments.
- Nigerian data-light mode, exam-season EOQ push, per-lecture EOQ banks (decision pending).

15 · Appendix: environments & operations

Render (backend)

`SUPABASE_URL · SUPABASE_PUBLISHABLE_KEY · SUPABASE_SECRET_KEY · DATABASE_URL (:6543) · DIRECT_URL (:5432) · CORS_ORIGIN · ADMIN_EMAILS · GEMINI_API_KEY · BREVO_API_KEY · EMAIL_FROM · VAPID_PUBLIC_KEY · VAPID_PRIVATE_KEY · VAPID_SUBJECT`

> `EMAIL_FROM` must equal a **verified Brevo sender** (the address confirmed during Brevo account setup); otherwise note/announcement mail fails with `brevo 400` even though the key is valid. Default fallback: `Unify Learn <notes@unify.learn>`.

Vercel (frontend)

`SUPABASE_URL · SUPABASE_PUBLISHABLE_KEY · API_URL · USE_BACKEND=1 · VITE_VAPID_PUBLIC_KEY` (optional override)

Supabase dashboard

Custom SMTP (Brevo SMTP key, verified sender, test mail) · URL config (`unify-virid.vercel.app/**`) · three branded templates (Confirm signup, Invite, Reset password) · Google provider + redirect URLs (live since v1.11.0) · Auth logs for signup 500s.

Repos & deploys

Full app: `knight-dev01/Unify` (Render builds backend, migrates DB). Frontend mirror: `DIBBLS/Unify` (Vercel) synced by `scripts/Sync-Frontend.ps1`. Every push: bump `APP_VERSION` → tsc both sides + production build → commit (version in message) → push → sync → Vercel auto-build; Render auto-build migrates.

16 · v1.7–v1.8 additions

Expiring share links (viral loop)

- Authors and admins mint `/s/:token` links per week (lifetimes 8/16/24h, default 24h); students consume, never mint (403 server-side).
- Recipients read the full week free — hero, topics, recalls, exam, read-aloud — then hit expiry and a join screen. Views counted; owner/admin can revoke.
- WhatsApp/Telegram unfurls via a serverless OG page (`/api/share/:token`) with title, topics, countdown and cover; edge-cached 5 minutes.

XP-gated perks

- Save/Print PDF unlocks at **300 XP** (30 completed topics); staff bypass. Gate truthfully re-checks (retry + refocus, never a fake 0).
- Framework in `lib/xp.ts` (`XP_GATES`) for future locks: longer share lifetimes, early content, themes.

Nigerian reading voice

- Read-aloud auto-picks an `en-NG` device voice when present (many Androids ship Google's English (Nigeria)), with a manual voice picker and per-device memory.
- Honest limit: the accent must exist in the OS/browser TTS engine — the app cannot invent one. Android users download it under Settings → Language → Text-to-speech.

Design toggle & fonts

- Profile → Appearance switches **Classic** (previous look, default) and **Story** (MEE editorial) across the whole app, headings included.
- Theme and design are per-device (localStorage), cross-tab synced, never synced across browsers.

Push permission prompt

- Designed bottom sheet ~3s after first entry (7-day snooze), plus a browser-denied guide (site settings → allow; iPhone Home-Screen note).

17 · v1.8.1–v1.11 bug-fix batch + new features

October 2026 testing found 14 confirmed bugs (7 critical, 7 important). All fixed, each verified with `tsc` both sides plus a production build. Full per-bug account in `docs/DEVELOPER-REPORT.md`.

Content pipeline hardened (v1.8.1–v1.8.5)

- Collaborator scoping server-side; app-wide MathJax renderer (KaTeX removed); real Storage diagrams; async chunked generation with retry; submit-at-end EOQ; course aliases (`ME 352 → MEE 352`) with cross-semester search; latest-version-only lists; Replace/Add/Cancel on occupied weeks.
- EOQ short banks auto-repair to exactly 10 questions (8 MCQ + 2 FITB); definition/formula cards render through MathJax with overlap-proof layout; quiz Check buttons styled.

Lecture structure (v1.9.0)

- Weeks hold **Lecture 1/2/3**, each with topics numbered from 1 and its own chapter tabs; the reader switches lectures without reload (`?c=`), with cross-lecture Back/Next. Progress, versions, resume and bookmarks are lecture-keyed. Studio/NoteBuilder carry lecture pickers; the AI prompt emits lecture groups. EOQ stays one week bank after the final lecture.

Dashboard, speed, Explore crash (v1.9.3–v1.9.4)

- Dashboard courses are a 2-column card grid (title, lecturer, weeks, topics done) with one-tap elective removal; catalog carries lecturer names; idle-time week-list warmup; Explore search-hook crash fixed.

Lecturer classes + offline-first (v1.10.0)

- **My Classes** (`/classes`): recurring weekly timetable per taught course (day/time/venue) plus a live roster (topics, quizzes, last activity). Students read class times on course pages.
- Offline foundation: progress/quiz/bookmark writes bank locally and flush on reconnect; per-week **Save offline** into a version-proof content cache; banner shows the banked count.

Google sign-in (v1.11.0)

- Continue with Google on both auth tabs (Supabase OAuth, PKCE); first-timers onboard with their Google name prefilled. Requires the provider + redirect URLs in the Supabase dashboard (see `SETUP.md`).

18 · v1.11.2–v1.24 change log

Compact digest of the 4–19 October pushes (full detail in the git log — every commit carries its version):

Version(s)	What shipped
v1.11.2–v1.11.3	Skeletons match their loads, nav active states, brand-kit cheap wins (focus states, nav pill, Playfair numbers).
v1.12.0–v1.12.1	Error visibility (global error toasts), WhatsApp support entry, push-driven notification bell, atomic replace on publish; wider desktop shell + quiz formula answers render.
v1.13.0–v1.13.1	Unify wordmark lockup app-wide, read-aloud speed control, new OG social card.
v1.14.0	Performance: vendor-split bundles, parallel dashboard loads, `me()` memoization, lazy media.
v1.15.0	Contributor role hardening (approval flow), avatars, caps groundwork, admin tables.
v1.16.0	Waiting room, quiz-math validation, auth lockup fix, OG refresh.
v1.17.0	Full catalog seed, admin scoping, staff onboarding trim.
v1.18.0	Server cap gate + audit, global toasts, new streak rule.
v1.19.0	Searchable assignment, cap audit UI, admin sidebar panel.
v1.20.0	Share fallback route, per-link OG authors, week author badges.
v1.21.0	Narrator audio track per topic.
v1.22.0	Word-like manual editing, review summaries, limits, score overlay; back buttons honor the return trail (incl. reader); publish fan-out paginated past 200/1000 caps; full 242-course seed verified.
v1.23.0	Tap-to-insert equation templates — no hand-typed LaTeX.
v1.24.0	Visual equation builder (zero LaTeX typing) + editable equation boxes, review beware checklist; PWA InstallBar (collapsible pill) + update-ready dot; inline Drop selects; assessment OG card; loader elapsed timer.

19 · v1.25–v1.26.7 (20 October batch)

- **Reader controls:** Focus mode (chrome hides only on reader routes), desktop zoom-in 100–150% targeting the content column only; first zoom-in enables Focus; both persist per device and survive note switches.
- **Overlay dropdowns:** the shared `Drop` component replaced every native select.

- **Math — never-raw (v1.26.4, resolved):** typeset failures were being swallowed, leaving raw `\frac` on screen; `typesetMath` now returns a boolean with a 6s deadline + retry, `Formula` shows glyphs from first paint and reveals only real typeset output, successful renders are **saved for offline**, and paragraphs fall back to glyph math. Equation editor gained a live RENDERED preview.
- **Email:** profile verify chip + Resend; note/announcement mail via Brevo API from `EMAIL_FROM` (verified sender required).
- **Avatars:** any-size upload downscaled on-device to a 256px JPEG; backend validates by magic bytes; SSO profile pictures backfilled; rejections surfaced verbatim.
- **Save PDF (v1.26.6):** print output cleaned — controls/install bar hidden, reader zoom reset, cards unbroken; documented in apps/web/README.
- **Delivery speed:** route-level code-splitting (main bundle 350→190 KB raw); jsDelivr preconnect for MathJax.
- **Docs:** READMEs, SETUP (Brevo/EMAIL_FROM + no-mail troubleshooting), UNIFY_MATHJAX_RULE, DEVELOPER-REPORT and this product doc all refreshed to v1.26.7.